

CHERVYANKA, Russian Federation

WMO: 293930

Lat: 57.654N

Lon: 99.542E

Elev: 221

StdP: 98.70

Time Zone: 8.00 (E08)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.1	(c) -40.3	(d) -44.3	(e) 0.0	(f) -42.3	(g) -41.9	(h) 0.1	(i) -39.6	(j) 8.8	(k) -11.1	(l) 7.7	(m) -10.4	(n) 0.1	(o) 250	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	29.8	18.7	27.9	18.2	26.0	17.4	20.4	26.6	19.3	25.4	18.4	24.1	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.6	22.8	17.2	12.6	21.4	16.2	11.8	20.7	59.9	26.1	56.2	25.5	53.2	24.4	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.5	(b) 7.2	(c) 6.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 8.5	(5) 7.2	(6) 6.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(4) 8.5	(5) 7.2
			WB	-44.2	22.6	2.3	0.6	-45.9	23.0	-47.2	23.4	-48.5	23.7	-50.2	24.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-22.6	-18.9	-8.2	1.5	7.5	15.6	17.8	15.3	6.7	-0.9	-13.4	-18.5	(6)	
	DBStd	15.40	9.81	8.18	7.45	4.66	4.50	4.16	2.84	3.36	4.28	5.40	8.93	9.81	(7)	
	HDD10.0	4794	1009	810	563	258	106	4	0	3	117	337	703	883	(8)	
	HDD18.3	7260	1268	1043	822	507	336	101	43	103	348	595	953	1141	(9)	
	CDD10.0	635	0	0	0	2	29	173	243	168	19	1	0	0	(10)	
	CDD18.3	58	0	0	0	0	1	19	28	9	0	0	0	0	(11)	
	CDH23.3	1099	0	0	0	0	23	455	399	214	7	0	0	0	(12)	
	CDH26.7	297	0	0	0	0	4	136	104	53	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	1.8	1.6	2.3	3.2	3.4	2.4	2.0	1.8	2.2	2.6	2.3	1.9	(14)	
(15) Precipitation	PrecAvg	311	12	10	8	14	29	34	56	59	33	21	21	18	(15)	
	PrecMax	458	29	19	22	26	64	63	142	129	73	43	50	34	(16)	
	PrecMin	197	5	1	0	1	8	8	10	7	9	2	7	3	(17)	
	PrecStd	59	6	5	5	7	13	16	31	29	16	11	10	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	0.7	9.3	19.1	26.3	32.0	31.9	30.8	24.6	15.0	6.6	1.3	(19)	
		MCWB	-1.9	-1.9	3.2	8.7	14.1	19.2	20.3	19.3	16.1	9.1	4.7	0.0	(20)	
	2%	DB	-3.1	-1.4	6.3	16.0	22.7	30.1	29.4	28.1	20.7	11.5	1.9	-1.6	(21)	
		MCWB	-4.0	-2.9	1.5	7.1	11.9	18.3	19.2	18.3	13.7	6.5	0.4	-2.8	(22)	
	5%	DB	-6.0	-3.5	4.2	12.8	20.1	28.2	27.5	25.7	17.1	8.8	0.3	-3.0	(23)	
		MCWB	-7.0	-5.0	0.4	5.7	11.0	17.7	18.6	17.3	11.9	5.2	-0.8	-3.8	(24)	
	10%	DB	-8.4	-6.1	2.5	10.1	17.3	26.1	25.5	23.6	14.2	6.3	-1.5	-4.9	(25)	
		MCWB	-9.1	-7.3	-0.6	4.6	9.4	16.9	17.9	16.6	10.5	3.5	-2.7	-5.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.6	-1.3	3.9	10.0	14.9	21.2	22.0	20.7	16.5	9.5	4.7	0.1	(27)	
		MCDB	-0.8	0.1	7.8	18.3	24.3	28.7	28.5	27.5	22.6	13.3	6.4	1.4	(28)	
	2%	WB	-4.0	-2.8	2.0	7.5	13.2	19.8	20.9	19.3	14.2	7.0	0.8	-2.6	(29)	
		MCDB	-2.9	-1.2	5.9	14.3	20.1	26.7	26.5	26.3	19.1	11.0	1.4	-1.9	(30)	
	5%	WB	-6.8	-4.9	0.7	6.2	11.7	18.7	19.7	18.2	12.6	5.3	-0.9	-3.8	(31)	
		MCDB	-5.9	-3.5	3.9	12.3	18.4	25.8	25.1	24.1	16.0	8.2	0.1	-3.0	(32)	
	10%	WB	-9.1	-7.3	-0.6	4.6	10.3	17.7	18.7	17.2	10.7	3.8	-2.6	-5.8	(33)	
		MCDB	-8.3	-6.1	2.4	9.6	16.7	24.7	23.8	22.1	14.0	6.1	-1.5	-4.8	(34)	
(35) Mean Daily Temperature Range	MDBR		10.8	14.3	14.3	11.2	12.6	14.8	12.9	13.3	11.0	8.6	9.4	9.6	(35)	
	5% DB	MCDBR	8.6	13.2	14.8	15.9	17.9	19.1	17.5	18.1	17.4	13.6	6.1	7.6	(36)	
		MCWBR	7.8	11.5	10.7	8.9	9.0	8.7	8.2	9.3	10.3	9.1	5.8	6.8	(37)	
	5% WB	MCDBR	8.7	12.6	12.9	14.1	15.4	16.4	13.7	15.8	14.7	12.4	6.6	6.8	(38)	
		MCWBR	7.9	11.1	9.8	8.8	8.3	7.9	7.2	8.6	9.2	8.8	6.1	6.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.284	0.311	0.340	0.374	0.400	0.436	0.453	0.420	0.357	0.340	0.300	0.270	(40)	
	taud		1.993	2.011	2.019	2.151	2.241	2.210	2.184	2.242	2.393	2.260	2.059	1.960	(41)	
	Ebn at Noon		523	692	786	830	833	804	778	778	783	670	499	411	(42)	
	Edh at Noon		61	94	123	127	125	131	132	116	86	74	56	46	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.68	1.64	3.07	4.55	5.14	5.86	5.43	4.31	2.64	1.28	0.66	0.41	(44)	
	RadStd		0.08	0.12	0.18	0.27	0.39	0.46	0.35	0.32	0.36	0.12	0.09	0.07	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page